

DA-3
Matrix Chain
Multiplication
Design And Analysis
Algorithm

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Ques → Given a chain of Matrix :-

$A_1 : 5 \times 4$ $A_4 : 5 \times 2$
 $A_2 : 4 \times 3$ $A_5 : 2 \times 4$
 $A_3 : 3 \times 5$

- Where A_i has Dimensions $P_{i-1} \times P_i$ fully parenthesized the product $A_1 A_2 \dots A_n$ in a way that minimize the member number of Scalar multiplication.

Solve- we will be using tabulation Method to solve this Question.

A_1	A_2	A_3	A_4	A_5
$P_0 \times P_1$	$P_1 \times P_2$	$P_2 \times P_3$	$P_3 \times P_4$	$P_4 \times P_5$
5×4	4×3	3×5	5×2	2×4

Row, Col = 5

	A1	A2	A3	A4	A5
A1	0	60	135	94	134
A2		0	60	54	86
A3			0	30	54
A4				0	40
A5					0

$$m(i, j) = m(i, k) + m(k+1, j) + p_{i-1} \times p_k \times p_j$$

$i=1, j=2$

① $m(1, 2) = m(1, 1) + m(2, 2) + 5 \times 4 \times 3 = 60$

② $m(2, 3) = m(2, 2) + m(3, 3) + 4 \times 3 \times 5 = 60$

③ $m(3, 4) = m(3, 3) + m(4, 4) + 3 \times 5 \times 2 = 30$

④ $m(4, 5) = m(4, 4) + m(5, 5) + 5 \times 2 \times 4 = 40$

⑤ $m(1, 3) = m(1, 1) + m(2, 3) + 5 \times 4 \times 5 = 160$

$k=2 \Rightarrow m(1, 2) + m(3, 3) + 5 \times 3 \times 5 = 135$

⑥ $m(2, 4)$

$k=2 \Rightarrow m(2, 2) + m(3, 4) + 4 \times 3 \times 2 = 54$

$k=3 \Rightarrow m(2, 3) + m(4, 4) + 4 \times 5 \times 2 = 100$

⑦ $m(3, 5)$

$k=3 \Rightarrow m(3, 3) + m(4, 5) + 3 \times 5 \times 4 = 100$

$k=4 \Rightarrow m(3, 4) + m(5, 5) + 3 \times 2 \times 4 = 54$

⑧ $m(1, 4)$

$k=1 \Rightarrow m(1, 1) + m(2, 4) + 5 \times 4 \times 2 = 94$

$k=2 \Rightarrow m(1, 2) + m(3, 4) + 5 \times 3 \times 2 = 120$

$k=3 \Rightarrow m(1, 3) + m(4, 4) + 5 \times 5 \times 2 = 185$

⑨ $m(2,5)$
 $k=2 \Rightarrow m(2,2) + m(3,5) + 4 \times 3 \times 4 = 102$
 $k=3 \Rightarrow m(2,3) + m(4,5) + 4 \times 5 \times 4 = 180$
 $k=4 \Rightarrow m(2,4) + m(5,5) + 4 \times 2 \times 4 = \textcircled{86}$

⑩ $m(7,5)$
 $k=1 \Rightarrow m(7,7) + m(2,5) + 5 \times 4 \times 4 = 166$
 $k=2 \Rightarrow m(7,2) + m(3,5) + 5 \times 3 \times 5 = 174$
 $k=3 \Rightarrow m(7,3) + m(4,5) + 5 \times 5 \times 4 = 275$
 $k=4 \Rightarrow m(7,4) + m(5,5) + 5 \times 2 \times 4 = \textcircled{134}$

Minimum Scalar Multiplication = 134

Optimal parenthesization: $A_1(A_2(A_3(A_4A_5)))$